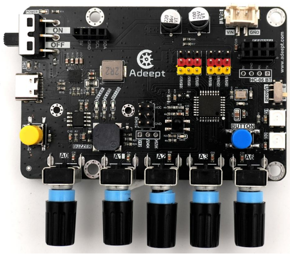
Lesson 2 Use the Buzzer to Plays Music

2.1 Overview

This lesson focuses on using the buzzer on the Adeept Robotic Arm Control Board to play various tones and a song. By following the steps and code examples provided, you can learn how to control the buzzer's frequency and duration to create different musical sounds.

2.2 Required Components

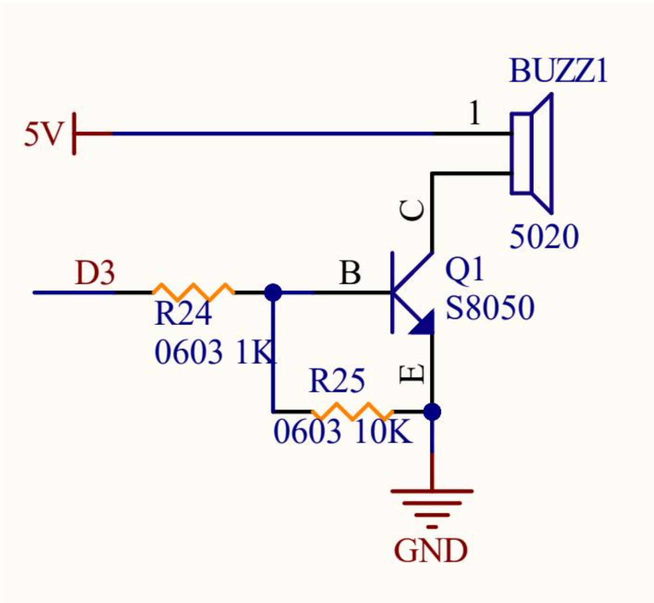
Components	Quantity	Picture
Adeept Robotic Arm Control Board	1	
Type-C USB Cable	1	

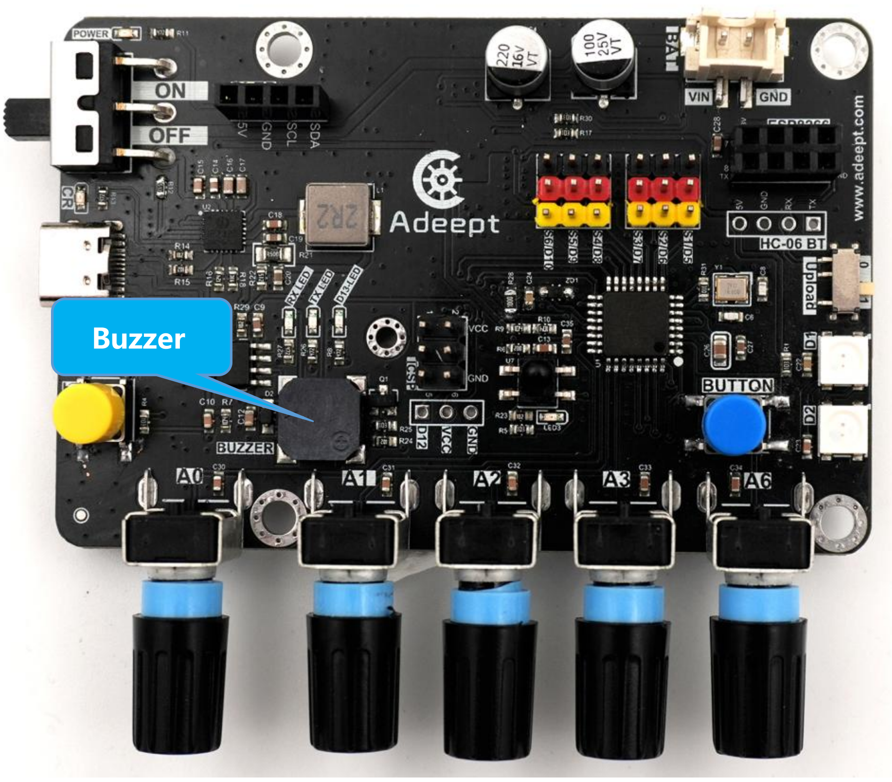
2.3 Principle Introduction

The Adeept Robotic Arm Control Board has a passive buzzer integrated into it. The buzzer is connected to pin D3 of the board. When an appropriate electrical signal is sent to this pin, the buzzer can produce sounds of different frequencies. The frequency of the electrical signal determines the pitch of the sound, and the duration of the signal controls how long the sound lasts.

2.4 Wiring Diagram

A passive buzzer is built into the Adeept Robotic Arm Control Board.



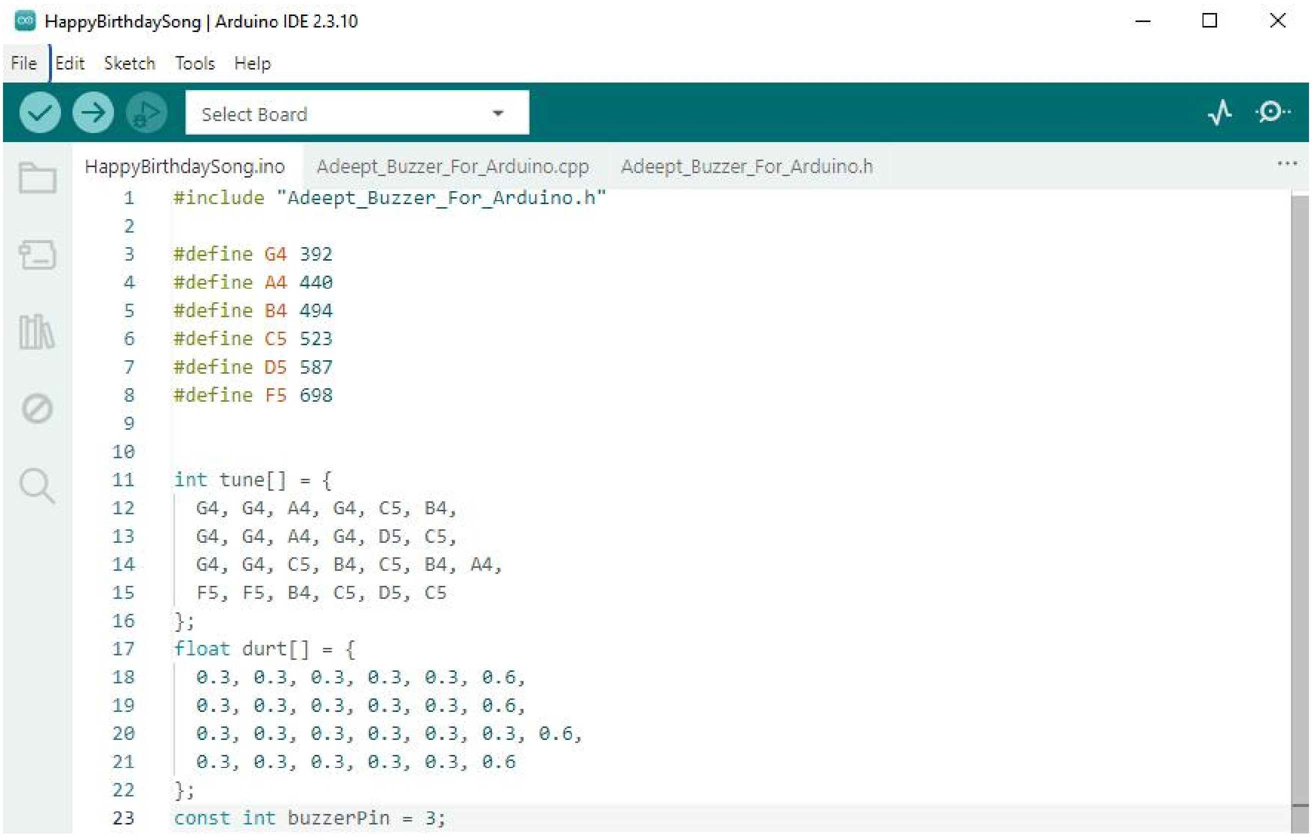


PINS of Adeept Robotic Arm Control Board	Buzzer
D3	Buzzer

2.5 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board with a USB cable.

2. Open "**02_Buzzer/HappyBirthdaySong**" folder in "**/Code**", double-click "**HappyBirthdaySong.ino**".

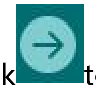


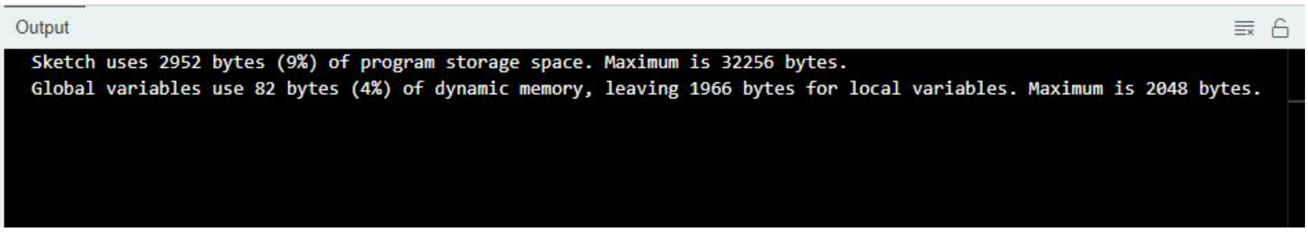
3. Select development board and serial port.

Board: **Tools--->Board--->Arduino AVR Boards--->Arduino Uno**

Port: **Tools --->Port--->COMx**

Note: The port number will be different in different computers.

4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.



5. After the program runs successfully, the onboard buzzer will play the "Happy Birthday" song.

2.6 Code

```
#include "Adeept_Buzzer_For_Arduino.h"

#define G4 392
#define A4 440
#define B4 494
#define C5 523
#define D5 587
#define F5 698


int tune[] = {
  G4, G4, A4, G4, C5, B4,
  G4, G4, A4, G4, D5, C5,
  G4, G4, C5, B4, C5, B4, A4,
  F5, F5, B4, C5, D5, C5
};
float durt[] = {
  0.3, 0.3, 0.3, 0.3, 0.3, 0.6,
  0.3, 0.3, 0.3, 0.3, 0.3, 0.6,
  0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.6,
  0.3, 0.3, 0.3, 0.3, 0.3, 0.6
};
const int buzzerPin = 3;

AdeeptBuzzer buzzer(buzzerPin);
int length = sizeof(tune) / sizeof(tune[0]);
void setup() {

}

void loop() {

  for (int i = 0; i < length; i++) {
    int duration = durt[i] * 1000;
    buzzer.playTone(tune[i], duration);
```

```
    delay(10);
  }
  delay(2000);
}
```

Complete code refer to Adeept_Buzzer_For_Arduino.cpp

```
#include "Adeept_Buzzer_For_Arduino.h"
#include <Arduino.h>

AdeeptBuzzer::AdeeptBuzzer(int pin) {
  buzzerPin = pin;
  pinMode(buzzerPin, OUTPUT);
}

void AdeeptBuzzer::playTone(int frequency, int duration) {
  tone(buzzerPin, frequency, duration);
  delay(duration);+
  noTone(buzzerPin);
}

```

Complete code refer to Adeept_Buzzer_For_Arduino.h

```
// Adeept_Buzzer_For_Arduino.h
#ifndef ADEEPT_BUZZER_FOR_ARDUINO_H
#define ADEEPT_BUZZER_FOR_ARDUINO_H

class AdeeptBuzzer {
  public:
    AdeeptBuzzer(int pin);
    void playTone(int frequency, int duration);
  private:
    int buzzerPin;
};

#endif
```